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INJECTION MOLDING MACHINE

Abstract

An injection molding machine includes: a mold clamping device including: a movable platen having a movable mold and a fixed platen having a fixed mold; and a toggle mechanism that moves the movable platen such that a mold device opens and closes, and disposes the movable platen in a mold clamping position; an injection device that injects molding material into cavity space between the movable mold in the mold clamping position and the fixed mold; and a control device that controls the mold clamping and injection devices. A cleaning process is performed, while the movable platen is at a burr-forming position and farther away from the fixed mold than at the mold clamping position, and the injection device injects the molding material into the mold device to mold an article with burrs. A toggle factor at the burr-forming position is a magnification factor of 1 or greater.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-030425, filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an injection molding machine.

2. Description of Related Art

[0003] In the technical field of injection molding machines, there is a technology whereby, using a hydraulic actuator, a distance between molds (a movable mold and a fixed mold) in a mold device is adjusted by inserting and removing a member for increasing the distance between the molds.

SUMMARY

[0004] According to one example of the present disclosure, an injection molding machine is provided. This injection molding machine includes: [0005] a mold clamping device including: [0006] a movable platen having a movable mold; [0007] a fixed platen having a fixed mold that faces the movable mold; and [0008] a toggle mechanism configured to move the movable platen in directions in which the mold device opens and closes, relative to the fixed platen, and dispose the movable platen in a mold clamping position; [0009] an injection device configured to inject a molding material into a cavity space formed between the movable mold disposed in the mold clamping position and the fixed mold; and [0010] a control device configured to control operation of the injection device and the mold clamping device, and, in this injection molding machine: [0011] the control device is configured to perform a cleaning process, in which, while the movable platen is moved to a burr-forming position where the movable mold is farther away from the fixed mold than at the mold clamping position, the injection device injects the molding material into the mold device to mold an article with burrs; and [0012] the toggle mechanism is configured such that a toggle factor at the burr-forming position is a magnification factor of 1 or greater. [0013] According to the above example of the present disclosure, burrs can be formed with a simple structure, so that the boundaries between the molds can be cleaned up efficiently.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a diagram that shows the state of an injection molding machine according to one embodiment of the present disclosure upon completion of mold opening;

[0015] FIG. 2 is a diagram that shows the state of the injection molding machine according to one embodiment of the present disclosure upon mold clamping;

[0016] FIG. 3 is a functional block diagram that shows examples of components in a control device;

[0017] FIG. 4A is a diagram that shows examples of steps in a molding cycle in injection molding;

[0018] FIG. 4B is a diagram that shows examples of steps in the molding cycle in a cleaning process;

[0019] FIG. 5A is a diagram that shows the state of a mold clamping device for injection molding;

[0020] FIG. 5B shows mold thickness adjustment before the cleaning process;
[0021] FIG. 5C shows a burr-forming position after a mold closing step in the cleaning process compared to a mold clamping position in injection molding;
[0022] FIG. 6 shows an enlarged view of a part of a toggle mechanism;
[0023] FIG. 7 is a first flowchart that shows how to perform the cleaning process; and
[0024] FIG. 8 is a second flowchart that shows how to perform the cleaning process.

DETAILED DESCRIPTION

[0025] When a distance is formed between molds as in the related art described above, the injection molding machine can deliberately form burrs between the molds by injection molding, and suck and remove the debris (resin mold deposits, gas adhesions, tar, etc.) that are produced at the boundaries between the molds.

[0026] If an injection molding machine is structured such that a distance-widening member such as the one mentioned above is provided in the mold device, inconveniences such as increased manufacturing costs, layout restrictions, and structural complications arise. In particular, if a distance-widening member is provided in the mold device every time the mold device is replaced, the cost is likely to increase significantly.

[0027] The present disclosure therefore aims to provide an injection molding machine that can form burrs with a simple structure, so that the boundaries between molds can be cleaned up efficiently.

[0028] Hereinafter, an embodiment of the present disclosure will be described with reference to the accompanying drawings. In the drawings, the same or corresponding components will be assigned the same or corresponding reference numerals, and the description thereof may be omitted.

(Injection Molding Machine)

[0029] FIG. 1 is a diagram that shows the state of an injection molding machine according to one embodiment of the present disclosure at the time of completion of mold opening. FIG. 2 is a diagram that shows the state of the injection molding machine according to the embodiment of the present disclosure at the time of mold clamping. In the present specification, the X-axis, Y-axis, and Z-axis directions are perpendicular to each other. The X-axis and Y-axis directions are horizontal directions, and the Z-axis direction is a vertical direction. In the event a mold clamping device **100** is a horizontal type, the X-axis direction is the mold opening/closing directions, and the Y-axis direction is the width direction of an injection molding machine **10**. The negative side in the Y-axis direction will be hereinafter referred to as an “operation side,” and the positive side in the Y-axis direction will be hereinafter referred to as a “side opposite to the operation side.”

[0030] As illustrated in FIG. 1 and FIG. 2, the injection molding machine **10** includes: a mold clamping device **100** that opens and closes a mold device **800**; an ejector device **200** that ejects a molded article that is molded by the mold device **800**; an injection device **300** that injects a molding material into the mold device **800**; a moving device **400** that allows the injection device **300** to move forward and backward (advance and retract) with respect to the mold device **800**; a control device **700** that controls components of the injection molding machine **10**; and a frame **900** that supports components of the injection molding machine **10**. The frame **900** includes a mold clamping device frame **910** that supports the mold clamping device **100** and an injection device frame **920** that supports the injection device **300**. The mold clamping device frame **910** and the injection device frame **920** are installed on a floor **2** via leveling adjusters **930**. The control device **700** is provided in the internal space of the injection device frame **920**. Now, each component of the injection molding machine **10** will be described below.

(Mold Clamping Device)

[0031] In the following description of the mold clamping device **100**, the direction in which a movable platen **120** moves at the time of mold closing (for example, the positive X-axis direction) is defined as toward the front, and the direction in which the movable platen **120** moves at the time of mold opening (for example, the negative X-axis direction) is defined as toward the rear.

[0032] The mold clamping device **100** performs mold closing, pressure increasing, mold clamping, pressure releasing, and mold opening, for the mold device **800**. The mold device **800** includes a fixed mold **810** and a movable mold **820**.

[0033] The mold clamping device **100** is, for example, a horizontal type, and the molds open and close in horizontal directions. The mold clamping device **100** includes: a fixed platen **110** to which the fixed mold **810** is attached; a movable platen **120** to which the movable mold **820** is attached; and a moving mechanism **102** that allows the movable platen **120** to move, with respect to the fixed platen **110**, in directions in which the molds open and close (hereinafter “mold opening/closing directions”).

[0034] The fixed platen **110** is fixed to the mold clamping device frame **910**. The fixed mold **810** is attached to the surface of the fixed platen **110** facing the movable platen **120**.

[0035] The movable platen **120** is disposed so as to be movable in mold opening/closing directions with respect to the mold clamping device frame **910**. A guide **101** for guiding the movable platen **120** is provided on the mold clamping device frame **910**. The movable mold **820** is attached to the surface of the movable platen **120** facing the fixed platen **110**.

[0036] The moving mechanism **102** moves the movable platen **120** forward and backward with respect to the fixed platen **110** to perform mold closing, pressure increasing, mold clamping, pressure releasing, and mold opening for the mold device **800**. The moving mechanism **102** includes: a toggle support **130** disposed at a distance from the fixed platen **110**; tie bars **140** that connect the fixed platen **110** and the toggle support **130**; a toggle mechanism **150** configured to allow the movable platen **120** to move in mold opening/closing directions with respect to the toggle support **130**; a mold clamping motor **160** configured to operate the toggle mechanism **150**; a motion conversion mechanism **170** configured to convert rotational motion of the mold clamping motor **160** into linear motion; and a mold thickness adjustment mechanism **180** configured to adjust the interval between the fixed platen **110** and the toggle support **130**.

[0037] The toggle support **130** is disposed at a distance from the fixed platen **110** and placed on the mold clamping device frame **910** so as to be movable in mold opening/closing directions. The toggle support **130** may be disposed so as to be movable along a guide that is provided on the mold clamping device frame **910**. The guide **101** for the movable platen **120** may also serve as the guide for the toggle support **130** as well.

[0038] In the present embodiment, although the fixed platen **110** is fixed to the mold clamping device frame **910** and the toggle support **130** is disposed so as to be movable in mold opening/closing directions with respect to the mold clamping device frame **910**, it is also possible to employ a structure in which the toggle support **130** is fixed to the mold clamping device frame **910** and the fixed platen **110** is disposed so as to be movable in mold opening/closing directions with respect to the mold clamping device frame **910**.

[0039] Each tie bar **140** connects the fixed platen **110** and the toggle support **130** with an interval **L** therebetween in mold opening/closing directions. Multiple (for example, four) tie bars **140** may be used. Multiple tie bars **140** may be arranged in parallel in mold opening/closing directions and extend in accordance with mold clamping force. At least one tie bar **140** may be provided with a tie bar strain detector **141** for detecting the strain of the tie bar **140**. The tie bar strain detector **141** sends signals representing detection results obtained thereat to the control device **700**. The detection results of the tie bar strain detector **141** is used, for example, to detect the mold clamping force or the like.

[0040] In the present embodiment, the tie bar strain detector **141** is used as a mold clamping force detector for detecting the mold clamping force, but the present disclosure is by no means limited to this. The mold clamping force detector is by no means limited to a strain gauge type, and may be a piezoelectric type, a capacitance type, a hydraulic type, an electromagnetic type, or the like, and the position where the mold clamping force detector is attached is by no means limited to the tie bar **140** either.

[0041] The toggle mechanism **150** is disposed between the movable platen **120** and the toggle support **130**, and allows the movable platen **120** to move in mold opening/closing directions with respect to the toggle support **130**. The structure of this toggle mechanism **150** will be described later in detail.

[0042] The mold clamping motor **160** is attached to the toggle support **130** and operates the toggle mechanism **150**. The mold clamping motor **160** moves the crosshead **151** forward and backward with respect to the toggle support **130**, so that a first link **152** and a second link **153** bend and extend, and the movable platen **120** moves forward and backward with respect to the toggle support **130**. The mold clamping motor **160** is directly connected to the motion conversion mechanism **170**, but may be connected to the motion conversion mechanism **170** via a belt, a pulley, or the like.

[0043] The motion conversion mechanism **170** converts the rotational motion of the mold clamping motor **160** into linear motion of the crosshead **151**. The motion conversion mechanism **170** includes a screw shaft and screw nuts that mesh with the screw shaft. Balls or rollers may be interposed between the screw shaft and the screw nuts.

[0044] The mold clamping device **100** performs a mold closing step, a pressure increasing step, a mold clamping step, a pressure releasing step, a mold opening step, and the like, under the control of the control device **700**.

[0045] In the mold closing step, the mold clamping motor **160** is driven to advance the crosshead **151** to a position where the molds are completely closed (hereinafter “mold closing completion position”) at a set moving speed, thereby allowing the movable platen **120** to move forward and the movable mold **820** to touch the fixed mold **810**. The position and the moving speed of the crosshead **151** are detected by using, for example, a mold clamping motor encoder **161**. The mold clamping motor encoder **161** detects the rotation of the mold clamping motor **160** and sends a signal representing the detection result to the control device **700**.

[0046] The crosshead position detector for detecting the position of the crosshead **151** and the crosshead moving speed detector for detecting the moving speed of the crosshead **151** are by no means limited to the mold clamping motor encoder **161**, and general detectors can be used. The movable platen position detector for detecting the position of the movable platen **120** and the movable platen moving speed detector for detecting the moving speed of the movable platen **120** are by no means limited to the mold clamping motor encoder **161**, and general detectors can be used.

[0047] In the pressure increasing step, a mold clamping force is generated by further driving the mold clamping motor **160** to allow the crosshead **151** to advance from the mold closing completion position to a mold clamping position.

[0048] In the mold clamping step, the mold clamping motor **160** is driven so that the position of the crosshead **151** is maintained at the mold clamping position. In the mold clamping step, the mold clamping force generated in the pressure increasing step is maintained. In the mold clamping step, a cavity space **801** (see FIG. 2) is formed between the movable mold **820** and the fixed mold **810**, and the injection device **300** fills the cavity space **801** with a liquid molding material. The filled molding material is solidified, and a molded article is obtained.

[0049] There may be more than one cavity space **801**. In this case, a number of molded articles can be obtained at the same time. An insert component may be disposed in a part of the cavity space **801**, and the molding material may fill another part of the cavity spaces **801**. By doing so, a molded article in which the insert component and the molding material are integrated is obtained.

[0050] In the pressure releasing step, the movable platen **120** is retracted by driving the mold clamping motor **160** to move the crosshead **151** back from the mold clamping position to the position where the mold starts opening (hereinafter “mold opening start position”), and the mold clamping force is reduced. The mold opening start position and the mold closing completion position may be the same position.

[0051] In the mold opening step, the movable platen **120** is retracted by driving the mold clamping

motor **160** to move the crosshead **151** back from the mold opening start position to the mold opening completion position at a set moving speed, and the movable mold **820** is separated from the fixed mold **810**. Subsequently, the ejector device **200** ejects the molded article from the movable mold **820**.

[0052] The setting conditions for the mold closing step, the pressure increasing step, and the mold clamping step are set collectively as a series of setting conditions. For example, the moving speed and positions of the crosshead **151** (including the mold closing start position, the moving speed switching position, the mold closing completion position, and the mold clamping position) in the mold closing step and the pressure increasing step, and the mold clamping force are set collectively as a series of setting conditions. The mold closing start position, the moving speed switching position, the mold closing completion position, and the mold clamping position are arranged in this order from the rear to the front, and represent, for example, the start point and end point of each section in which the moving speed is set. The moving speed is set in each section. The number of moving speed switching positions may be one or more. It is equally possible not to set a moving speed switching position. Only one of the mold clamping position and the mold clamping force need be set.

[0053] The setting conditions for the pressure releasing step and the mold opening step are set in the same manner. For example, the moving speed and position of the crosshead **151** (including the mold opening start position, the moving speed switching position, and the mold opening completion position) in the pressure releasing step and the mold opening step are set collectively as a series of setting conditions. The mold opening start position, the moving speed switching position, and the mold opening completion position are arranged in this order from the front to the rear, and represent, for example, the start point and end point of each section in which the moving speed is set. The moving speed is set in each section. The number of moving speed switching positions may be one or more. It is equally possible not to set a moving speed switching position. The mold opening start position and the mold closing completion position may be the same position. The mold opening completion position and the mold closing start position may be the same position.

[0054] Instead of the moving speed and positions of the crosshead **151**, the moving speed and positions of the movable platen **120** may be set. Instead of setting the crosshead's position (for example, the mold clamping position) or setting the movable platen's position, the mold clamping force may be set.

[0055] In the event the thickness of the mold device **800** changes when, for example, the mold device **800** is replaced, the temperature of the mold device **800** changes, and so on, the mold's thickness is adjusted such that a predetermined mold clamping force is obtained at the time of mold clamping. In the mold thickness adjustment, for example, the interval L between the fixed platen **110** and the toggle support **130** is adjusted such that the link angle θ of the toggle mechanism **150** becomes a predetermined angle at the time the movable mold **820** touches the fixed mold **810** (or at the time of "mold touch").

[0056] The mold clamping device **100** includes a mold thickness adjustment mechanism **180**. The mold thickness adjustment mechanism **180** adjusts the mold's thickness by adjusting the interval L between the fixed platen **110** and the toggle support **130**. The mold thickness adjustment is performed, for example, at a timing between the end of one molding cycle and the start of the next molding cycle. The mold thickness adjustment mechanism **180** includes, for example, a screw shaft **181** formed at a rear end portion of each tie bar **140**, a screw nut **182** held by the toggle support **130** so as to be rotatable yet not movable forward and backward, and a mold thickness adjustment motor **183** that rotates the screw nut **182** screwed to the screw shaft **181**.

[0057] The screw shaft **181** and the screw nut **182** are provided for each tie bar **140**. The rotational driving force of the mold thickness adjustment motor **183** may be transmitted to a number of screw nuts **182** via a rotational driving force transmission part **185**. Multiple screw nuts **182** can be

rotated synchronously. It is also possible to make multiple screw nuts **182** rotate individually by changing the transmission path from the rotational driving force transmission part **185**.

[0058] The rotational driving force transmission part **185** is configured by, for example, gears. In this case, a driven gear is formed on the outer periphery of each screw nut **182**. A driving gear is attached to the output shaft of the mold thickness adjustment motor **183**, and an intermediate gear that meshes with multiple driven gears and the driving gear is rotatably held at the center of the toggle support **130**. The rotational driving force transmission part **185** may be configured by belts, pulleys, or the like, instead of gears.

[0059] The operation of the mold thickness adjustment mechanism **180** is controlled by the control device **700**. The control device **700** drives the mold thickness adjustment motor **183** to rotate the screw nuts **182**. As a result of this, the position of the toggle support **130** with respect to the tie bar **140** is adjusted, and the interval L between the fixed platen **110** and the toggle support **130** is adjusted. Multiple mold thickness adjustment mechanisms may be used in combination.

[0060] The interval L is detected by using the mold thickness adjustment motor encoder **184**. The mold thickness adjustment motor encoder **184** detects the amount of rotation and the direction of rotation of the mold thickness adjustment motor **183** and sends a signal representing the detection result to the control device **700**. The detection result of the mold thickness adjustment motor encoder **184** is used to monitor and control the position of the toggle support **130**, the interval L, and so forth. The toggle support position detector for detecting the position of the toggle support **130** and the interval detector for detecting the interval L are by no means limited to the mold thickness adjustment motor encoder **184**, and general detectors can be used as well.

[0061] The mold clamping device **100** may include a mold temperature adjuster that controls the temperature of the mold device **800**. The mold device **800** has a flow path for a temperature adjusting medium. The mold temperature adjuster adjusts the temperature of the mold device **800** by adjusting the temperature of the temperature adjusting medium supplied to the flow path in the mold device **800**.

[0062] The mold clamping device **100** of the present embodiment is a horizontal type, in which the mold opens and closes horizontally, but may be a vertical type in which the mold opens and closes vertically.

[0063] The mold clamping device **100** of the present embodiment includes the mold clamping motor **160** as a drive part but may include a hydraulic cylinder instead of the mold clamping motor **160**. Also, the mold clamping device **100** may include a linear motor for opening and closing the mold and an electromagnet for clamping the mold.

(Ejector Device)

[0064] In the following description of the ejector device **200**, similarly to the description of the mold clamping device **100**, the direction in which the movable platen **120** moves at the time of mold closing (for example, the positive X-axis direction) is defined as toward the front, and the direction in which the movable platen **120** moves at the time of mold opening (for example, the negative X-axis direction) is defined as toward the rear.

[0065] The ejector device **200** is attached to the movable platen **120** and moves forward and backward together with the movable platen **120**. The ejector device **200** includes ejector rods **210** that eject the molded article from the mold device **800**, and a drive mechanism **220** that moves the ejector rods **210** in directions in which the movable platen **120** moves (the positive and negative X-axis directions).

[0066] Each ejector rod **210** is disposed in a through-hole of the movable platen **120** so as to be movable forward and backward. The front end portion of each ejector rod **210** is in contact with an ejector plate **826** of the movable mold **820**. The front end portion of each ejector rod **210** may be connected with the ejector plate **826** or need not be connected with the ejector plate **826**.

[0067] The drive mechanism **220** includes, for example, an ejector motor and a motion conversion mechanism that converts the rotational motion of the ejector motor into linear motion of the ejector

rods **210**. The motion conversion mechanism include a screw shaft and screw nuts that mesh with the screw shaft. Balls or rollers may be interposed between the screw shaft and the screw nuts.

[0068] The ejector device **200** performs the ejection process under the control of the control device **700**. In the ejection step, the ejector rods **210** are moved forward from the standby positions to the ejection positions at a set moving speed, allowing the ejector plate **826** to move forward and eject the molded article. Subsequently, the ejector motor is driven to retract the ejector rods **210** at a set moving speed, and the ejector plate **826** is retracted to resume its original standby position.

[0069] The position and moving speed of the ejector rods **210** are detected by using, for example, an ejector motor encoder. The ejector motor encoder detects the rotation of the ejector motor and sends a signal representing the detection result to the control device **700**. The ejector rod position detector for detecting the positions of the ejector rods **210** and the ejector rod moving speed detector for detecting the moving speed of the ejector rods **210** are by no means limited to an ejector motor encoder, and general detectors can be used as well.

(Injection Device)

[0070] In the following description of the injection device **300**, unlike the description of the mold clamping device **100** and the description of the ejector device **200**, the direction in which the screw **330** moves at the time of filling (for example, the negative X-axis direction) is defined as toward the front, and the direction in which the screw **330** moves at the time of measuring (for example, the positive X-axis direction) is defined as toward the rear.

[0071] The injection device **300** is installed on a slide base **301**, and the slide base **301** is disposed so as to be movable forward and backward with respect to an injection device frame **920**. The injection device **300** is disposed so as to be movable forward and backward with respect to the mold device **800**. The injection device **300** touches the mold device **800** and fills the cavity space **801** inside the mold device **800** with the molding material. The injection device **300** includes, for example, a cylinder **310** that heats the molding material, a nozzle **320** provided at a front end portion of the cylinder **310**, a screw **330** disposed in the cylinder **310** so as to be movable forward and backward and rotatable, a measuring motor **340** that rotates the screw **330**, an injection motor **350** that moves the screw **330** forward and backward, and a load detector **360** that detects the load transmitted between the injection motor **350** and the screw **330**.

[0072] The cylinder **310** heats the molding material supplied from a supply port **311** to the inside. The molding material includes, for example, a resin. The molding material is formed in a pellet shape, for example, and is supplied to the supply port **311** in a solid state. The supply port **311** is formed in a rear portion of the cylinder **310**. A cooler **312**, such as a water-cooled cylinder or the like, is provided on the outer periphery of the rear portion of the cylinder **310**. A first heater **313**, such as a band heater or the like, and a first temperature detector **314** are provided on the outer periphery of the cylinder **310** in front of the cooler **312**.

[0073] The cylinder **310** is divided into a number of zones in the axial direction (for example, the X-axis direction) of the cylinder **310**. The first heater **313** and the first temperature detector **314** are provided for each zone. A set temperature is set for each zone, and the control device **700** controls the first heater **313** such that the temperature detected by the first temperature detector **314** reaches the set temperature.

[0074] The nozzle **320** is provided at a front end portion of the cylinder **310** and is pressed against the mold device **800**. A second heater **323** and a second temperature detector **324** are provided on the outer periphery of the nozzle **320**. The control device **700** controls the second heater **323** such that the temperature detected by the nozzle **320** reaches the set temperature.

[0075] The screw **330** is disposed in the cylinder **310** so as to be rotatable and movable forward and backward. When the screw **330** is rotated, the molding material is pushed forward along the spiral groove of the screw **330**. The molding material is pushed forward and gradually melted by the heat from the cylinder **310**. The liquid molding material is pushed to the front of the screw **330** and accumulated in a front portion of the cylinder **310**, and the screw **330** is retracted. Subsequently,

when the screw **330** is advanced, the liquid molding material accumulated in front of the screw **330** is injected from the nozzle **320** and fills the mold device **800**.

[0076] A backflow prevention ring **331** is attached to a front portion of the screw **330** so as to be movable forward and backward. The backflow prevention ring **331** serves as a backflow prevention valve that prevents backflow of the molding material from the front to the rear of the screw **330** when the screw **330** is pushed forward.

[0077] When the screw **330** is advanced, the backflow prevention ring **331** is pushed backward by the pressure of the molding material in front of the screw **330** and retracted, relative to the screw **330**, to a closure position (see FIG. 2) at which the backflow prevention ring **331** blocks the flow path of the molding material. This prevents the molding material accumulated in front of the screw **330** from flowing backward.

[0078] On the other hand, when the screw **330** is rotated, the backflow prevention ring **331** is pushed forward by the pressure of the molding material pushed forward along the spiral groove of the screw **330** and moves forward, relative to the screw **330**, to a release position (see FIG. 1) at which the backflow prevention ring **331** opens the flow path of the molding material. Thus, the molding material is pushed to the front of the screw **330**.

[0079] The backflow prevention ring **331** may be either a co-rotation type that rotates together with the screw **330** or a non-co-rotation type that does not rotate with the screw **330**.

[0080] The injection device **300** may include a drive source that moves the backflow prevention ring **331** forward and backward between the release position and the closure position with respect to the screw **330**.

[0081] The measuring motor **340** rotates the screw **330**. The drive source for rotating the screw **330** is by no means limited to the measuring motor **340**, and may be, for example, a hydraulic pump.

[0082] The injection motor **350** moves the screw **330** forward and backward. Between the injection motor **350** and the screw **330**, a motion conversion mechanism or the like for converting the rotational motion of the injection motor **350** into linear motion of the screw **330** is provided. The motion conversion mechanism includes, for example, a screw shaft and screw nuts that mesh with the screw shaft. Balls, rollers, or the like may be provided between the screw shaft and the screw nuts. The drive source for advancing and retracting the screw **330** is by no means limited to the injection motor **350**, and may be, for example, a hydraulic cylinder.

[0083] The load detector **360** detects the load transmitted between the injection motor **350** and the screw **330**. The detected load is converted into a pressure by the control device **700**. The load detector **360** is provided in a load transmission path between the injection motor **350** and the screw **330**, and detects the load that acts on the load detector **360**.

[0084] The load detector **360** sends a signal representing the detected load to the control device **700**. The load that is detected by the load detector **360** is converted into a pressure that acts between the screw **330** and the molding material, and is used to control or monitor the pressure that the screw **330** receives from the molding material, the back pressure upon the screw **330**, the pressure that acts on the molding material from the screw **330**, and the like.

[0085] The pressure detector for detecting the pressure of the molding material is by no means limited to the load detector **360**, and a general pressure detector can be used. For example, a nozzle pressure sensor or a mold internal pressure sensor may be used. A nozzle pressure sensor may be provided in the nozzle **320**. A mold internal pressure sensor may be provided inside the mold device **800**.

[0086] The injection device **300** performs the measurement step, a filling step, a maintenance step, and the like under the control of the control device **700**. The filling step and the maintenance step may be collectively referred to as "injection step."

[0087] In the measurement step, the measuring motor **340** is driven to rotate the screw **330** at a set rotation speed, and the molding material is pushed forward along the spiral groove of the screw **330**. As a result of this, the molding material is gradually melted. As the liquid molding material is

pushed to the front of the screw **330** and accumulated in a front portion of the cylinder **310**, and the screw **330** is retracted. The rotation speed of the screw **330** is detected by using, for example, the measuring motor encoder **341**. The measuring motor encoder **341** detects the rotation of the measuring motor **340** and sends a signal representing the detection result to the control device **700**. The screw rotational speed detector for detecting the rotational speed of the screw **330** is by no means limited to the measuring motor encoder **341**, and a general detector can be used as well. [0088] In the measurement step, in order to prevent or substantially prevent the screw **330** from moving backward suddenly, the injection motor **350** may be driven to apply a predetermined back pressure to the screw **330**. The back pressure to the screw **330** is detected by using, for example, the load detector **360**. When the screw **330** moves backward to a measuring completion position and a predetermined amount of molding material is accumulated in front of the screw **330**, the measurement step is completed.

[0089] The positions and rotation speed of the screw **330** in the measurement step are set collectively as a series of setting conditions. For example, a measurement start position, a rotational speed switching position, and a measuring completion position are set. These positions are arranged in this order from the front to the rear, and represent, for example, the start point and the end point of each section in which the rotation speed is set. The rotation speed is set in each section. The number of rotation speed switching positions may be one or more. The rotation speed switching position need not be set. Furthermore, a back pressure is set in each section.

[0090] In the filling step, the injection motor **350** is driven to move the screw **330** forward at a set moving speed, and the liquid molding material accumulated in front of the screw **330** is filled in the cavity space **801** in the mold device **800**. The position and the moving speed of the screw **330** are detected by using, for example, the injection motor encoder **351**. The injection motor encoder **351** detects the rotation of the injection motor **350** and sends a signal representing the detection result to the control device **700**. When the position of the screw **330** reaches a set position, the filling step switches to the maintenance step (known as “V/P switchover”). The position where the V/P switchover takes place is also referred to as a “V/P switchover position.” The set moving speed of the screw **330** may be changed according to the position of the screw **330**, time, and so forth.

[0091] The positions and moving speed of the screw **330** in the filling step are set collectively as a series of setting conditions. For example, a filling start position (also referred to as an “injection start position”), a moving speed switching position, and a V/P switchover position are set. These positions are arranged in this order from the rear to the front, and represent, for example, the start point and the end point of each section in which the rotation speed is set. The moving speed is set in each section. The number of moving speed switching positions may be one or more. It is equally possible not to set a moving speed switching position.

[0092] An upper limit value for the pressure of the screw **330** is set in each section in which the moving speed of the screw **330** is set. The pressure of the screw **330** is detected by the load detector **360**. In the event the pressure of the screw **330** is lower than or equal to the set pressure, the screw **330** moves forward at the set moving speed. On the other hand, in the event the pressure of the screw **330** exceeds the set pressure, for the purpose of protecting the mold, the screw **330** moves forward at a lower moving speed than the set moving speed so that the pressure of the screw **330** becomes lower than or equal to the set pressure.

[0093] Note that, after the position of the screw **330** reaches the V/P switchover position in the filling step, the screw **330** may be temporarily stopped at the V/P switchover position, and then a V/P switchover may take place. Shortly before the V/P switchover, it is also possible to move the screw **330** forward or backward at a very low speed, instead of stopping the screw **330**. The screw position detector for detecting the position of the screw **330** and the screw moving speed detector for detecting the moving speed of the screw **330** are by no means limited to the injection motor encoder **351**, and general detectors can be used as well.

[0094] In the maintenance step, the injection motor **350** is driven to push the screw **330** forward,

the pressure of the molding material at a front end portion of the screw **330** (hereinafter also referred to as “maintenance pressure”) is maintained at the set pressure, and the molding material that remains in the cylinder **310** is pushed toward the mold device **800**. The molding material that is short due to cooling shrinkage in the mold device **800** can be thus replenished. The maintenance pressure is detected by using, for example, the load detector **360**. The set value of the maintenance pressure may be changed according to the length of time elapsed from the start of the maintenance step. Multiple maintenance pressures and multiple durations of time for maintaining the maintenance pressures in the maintenance step may be set, and may be set collectively as a series of setting conditions.

[0095] In the maintenance step, the molding material in the cavity space **801** in the mold device **800** is gradually cooled down. When the maintenance step is completed, the inlet of the cavity space **801** is closed by the solidified molding material. This state is referred to as a “gate seal,” and a backflow of the molding material from the cavity space **801** is prevented. After the maintenance step is performed, the cooling step is started. In the cooling step, the molding material in the cavity space **801** is solidified. In order to shorten the molding cycle time, the measurement step may be performed during the cooling step.

[0096] The injection device **300** of the present embodiment is an in-line screw type, but may be a pre-plasticization type or the like. A pre-plasticizing injection device supplies the molding material melted in a plasticizing cylinder to an injection cylinder, and injects the molding material from the injection cylinder into the mold device. In the plasticizing cylinder, a screw that is able to rotate and unable to move forward and backward may be disposed, or a screw that is able to rotate and move forward and backward may be disposed. On the other hand, a plunger is disposed in the injection cylinder so as to be movable forward and backward.

[0097] The injection device **300** of the present embodiment is a horizontal type in which the axial direction of the cylinder **310** is horizontal, but may be a vertical type in which the axial direction of the cylinder **310** is vertical. A vertical or horizontal mold clamping device may be combined with a vertical injection device **300**. Similarly, a vertical or horizontal mold clamping device may be combined with a horizontal injection device **300**.

(Moving Device)

[0098] In the following description of the moving device **400**, similarly to the description of the injection device **300**, the direction in which the screw **330** moves at the time of filling (for example, the negative X-axis direction) is defined as toward the front, and the direction in which the screw **330** moves at the time of measuring (for example, the positive X-axis direction) is defined as toward the rear.

[0099] The moving device **400** advances and retracts the injection device **300** with respect to the mold device **800**. The moving device **400** presses the nozzle **320** against the mold device **800** to generate a nozzle touch pressure. The moving device **400** includes a hydraulic pump **410**, a motor **420** that serves as a drive source, a hydraulic cylinder **430** that serves as a hydraulic actuator, and so forth.

[0100] The hydraulic pump **410** has a first port **411** and a second port **412**. The hydraulic pump **410** is a pump that can rotate in both directions, and generates a hydraulic pressure by switching the rotation direction of the motor **420** and sucking in the working liquid (for example, oil) from one of the first port **411** and the second port **412** and discharging the liquid from the other port. The hydraulic pump **410** can also suck in the working liquid from a tank and discharge the working liquid from one of the first port **411** and the second port **412**.

[0101] The motor **420** operates the hydraulic pump **410**. The motor **420** drives the hydraulic pump **410** in a rotational direction and with a rotational torque corresponding to a control signal from the control device **700**. The motor **420** may be an electric motor or an electric servo motor.

[0102] The hydraulic cylinder **430** includes a cylinder main body **431**, a piston **432**, and a piston rod **433**. The cylinder main body **431** is fixed to the injection device **300**. The piston **432** divides

the inside of the cylinder main body **431** into a front chamber **435**, which serves as a first chamber, and a rear chamber **436**, which serves as a second chamber. The piston rod **433** is fixed to the fixed platen **110**.

[0103] The front chamber **435** of the hydraulic cylinder **430** is connected to the first port **411** of the hydraulic pump **410** via a first flow path **401**. The working liquid discharged from the first port **411** is supplied to the front chamber **435** via the first flow path **401**, so that the injection device **300** is pushed forward. The injection device **300** is advanced, and the nozzle **320** is pressed against the fixed mold **810**. The front chamber **435** functions as a pressure chamber that generates a nozzle touch pressure for the nozzle **320** by the pressure of the working liquid supplied from the hydraulic pump **410**.

[0104] The rear chamber **436** of the hydraulic cylinder **430** is connected to the second port **412** of the hydraulic pump **410** via a second flow path **402**. The working liquid discharged from the second port **412** is supplied to the rear chamber **436** of the hydraulic cylinder **430** via the second flow path **402**, so that the injection device **300** is pushed backward. The injection device **300** is retracted, and the nozzle **320** is separated from the fixed mold **810**.

[0105] In the present embodiment, the moving device **400** includes the hydraulic cylinder **430**, but the present disclosure is by no means limited to this example. For example, instead of the hydraulic cylinder **430**, an electric motor and a motion conversion mechanism that converts the rotational motion of the electric motor into linear motion of the injection device **300** may be used.

(Control Device)

[0106] The control device **700** is configured by, for example, a computer, and includes a central processing unit (CPU) **701**, a storage medium **702** such as a memory, an input interface **703**, and an output interface **704** as illustrated in FIG. 1 and FIG. 2. The control device **700** performs various controls by causing the CPU **701** to execute programs stored in the storage medium **702**. The control device **700** receives a signal from an external device through the input interface **703**, and transmits a signal to the external device through the output interface **704**.

[0107] The control device **700** repeats performing the measurement step, the mold closing step, the pressure increasing step, the mold clamping step, the filling step, the maintenance step, the cooling step, the pressure releasing step, the mold opening step, the ejection step, and the like, thereby manufacturing molded articles on a continuous basis (see FIG. 4). A series of operations for obtaining a molded article, for example, the operation from the start of the measurement step to the start of the next measurement step is also referred to as a “shot” or a “molding cycle.” The time required for one shot is referred to as “molding cycle time” or “cycle time.”

[0108] One molding cycle includes, for example, a measurement step, a mold closing step, a pressure increasing step, a mold clamping step, a filling step, a maintenance step, a cooling step, a pressure releasing step, a mold opening step, and an ejection step in this order. The order here is the order of in which each step starts. The filling step, the maintenance step, and the cooling step are performed during the mold clamping step. The start of the mold clamping step may match the start of the filling step. The completion of the pressure releasing step match the start of the mold opening step.

[0109] In order to shorten the molding cycle time, multiple steps may be performed at the same time. For example, the measurement step may be performed during the cooling step of the previous molding cycle, or may be performed during the mold clamping step. In this case, the mold closing step may be performed at the beginning of the molding cycle. The filling step may be started during the mold closing step. The ejection step may be started during the mold opening step. In the event an opening/closing valve that opens and closes the flow path of the nozzle **320** is provided, the mold opening step may be started during the measurement step. This is because even if the mold opening step is started during the measurement step, the molding material does not leak from the nozzle **320** as long as the opening/closing valve closes the flow path of the nozzle **320**.

[0110] One molding cycle may include steps other than the measurement step, the mold closing

step, the pressure increasing step, the mold clamping step, the filling step, the maintenance step, the cooling step, the pressure releasing step, the mold opening step, and the ejection step.

[0111] For example, after the maintenance step is completed, before the measurement step is started, a pre-measurement suck-back step of retracting the screw **330** to a predetermined measurement start position may be performed. This reduces the pressure of the molding material accumulated in front of the screw **330** before the measurement step is started, thereby preventing the screw **330** from retracting suddenly when the measurement step is started.

[0112] After the measurement step is completed, before the filling step is started, a post-measuring suck-back step may be performed to retract the screw **330** to a preset filling start position (also referred to as “injection start position”). This reduces the pressure of the molding material accumulated in front of the screw **330** before the filling step is started, and a leakage of the molding material from the nozzle **320** before the filling step is started can be prevented.

[0113] The control device **700** is connected with an operation device **750** which receives operational inputs from the user, and with a display device **760**, on which a screen is displayed. The operation device **750** and the display device **760** may be configured by, for example, a touch panel **770**, and may be integrated together. The touch panel **770**, which serves as the display device **760**, displays a screen under the control of the control device **700**. For example, information, such as the settings of the injection molding machine **10** and the current state of the injection molding machine **10**, may be displayed on the screen of the touch panel **770**. On the screen of the touch panel **770**, an operation part, such as buttons or an input field for receiving operational inputs from the user, may be displayed. The touch panel **770**, which serves as the operation device **750**, detects operational inputs from the user on the screen, and outputs signals corresponding to the operational inputs to the control device **700**. Thus, for example, the user can perform setting (including inputting setting values) of the injection molding machine **10** by operating the operation part provided on the screen while checking information displayed on the screen. The user operates the operation part provided on the screen, and the operation of the injection molding machine **10** is thus controlled according to operations made on the operation part. The operation of the injection molding machine **10** may refer to, for example, the operation (including stopping) of the mold clamping device **100**, the ejector device **200**, the injection device **300**, the moving device **400**, and so forth. The operation of the injection molding machine **10** may refer to switching of the screen displayed on the touch panel **770** as the display device **760**.

[0114] Note that the operation device **750** and the display device **760** of the present embodiment have been described as being integrated together as the touch panel **770**, but may be provided separately. Also, multiple operation devices **750** may be provided. The operation device **750** and the display device **760** are disposed on the operation side (negative Y-axis direction) of the mold clamping device **100** (more specifically, the fixed platen **110**).

(Details of Control Device)

[0115] Next, examples of the components of the control device **700** will be described with reference to FIG. **3**. Note that the functional blocks illustrated in FIG. **3** are conceptual and do not necessarily have to be provided physically as illustrated in FIG. **3**. Some or all of the functional blocks may be functionally or physically distributed or integrated together in any units as desired. All or any part of the processing functions performed by the functional blocks may be implemented by programs executed by the CPU or may be implemented as hardware by wired logic.

[0116] As shown in FIG. **3**, the control device **700** includes, for example, a mold clamping control part **711**, an ejector control part **712**, an injection control part **713**, a measurement control part **714**, a display control part **715**, and an input acquiring part **716**. The mold clamping control part **711** controls the mold clamping device **100**, and performs the mold closing step, pressure increasing step, mold clamping step, pressure releasing step, and mold opening step as shown in FIG. **4**. The ejector control part **712** controls the ejector device **200** and performs the ejection step. The injection control part **713** controls the injection drive source of the injection device **300** and

performs the injection step. The injection drive source is, for example, the injection motor **350**, but may also be a hydraulic cylinder or the like. The injection step includes a filling step and a maintenance step. The injection step is performed during the mold clamping step. The measurement control part **714** controls the measuring drive source of the injection device **300** and carries out the measurement step. The measuring drive source is, for example, a measuring motor **340**, but may also be a hydraulic pump or the like. The measurement step is performed during the cooling step.

[0117] The filling step is a process of controlling the injection drive source such that the actual value of the moving speed of an injection member provided inside the cylinder **310** becomes the set value. The filling step is a process of filling the inside of the mold device **800** with the liquid molding material (for example, resin) accumulated in front of the injection member by moving the injection member forward. The injection member is, for example, the screw **330**, but may also be a plunger.

[0118] The moving speed of the injection member is detected using a speed detector. The speed detector is, for example, the injection motor encoder **351**. In the filling step, the pressure that acts on the molding material from the injection member increases as the injection member moves forward. The filling step may include, shortly before the maintenance step, a step of temporarily stopping the injection member or a step of retracting the injection member.

[0119] The maintenance step is a process of controlling the injection drive source such that the actual value of the pressure that acts on the molding material from the injection member becomes a set value. The maintenance step is a process of pushing the injection member forward to replenish the molding material that is short due to cooling contraction in the mold device **800**. The pressure is detected using a pressure detector such as the load detector **360**. As for the pressure detector, a nozzle pressure sensor or a mold internal pressure sensor may be used.

[0120] Also, the display control part **715** of the control device **700** transmits display screen information related to injection molding, for example, before injection molding is started, in each step during injection molding, and after injection molding is completed, and controls the display device **760** to display each piece of information. Multiple display screens are prepared, so that the display control part **715** allows multiple screens to be switched between and displayed, laid over one another and displayed, and so forth.

[0121] Furthermore, the input acquiring part **716** of the control device **700** acquires information about the details of operations the user performs on the operation device **750**, based on the display screen displayed on the display device **760**. For example, when the settings of the injection molding are changed by the user, the input acquiring part **716** stores the settings in the storage medium **702**.

(Cleaning Process)

[0122] In the injection molding machine **10**, during injection molding, debris (resin mold deposits, gas adhesions, tar, etc.) are sometimes produced at the boundaries between the fixed mold **810** and the movable mold **820** of the mold device **800**. With conventional injection molding machines, a worker performs cleaning to remove such debris by, for example, wiping the boundaries between the molds with a cloth soaked in a cleaning agent such as alcohol. This cleaning is performed when injection molding is repeated a set number of times, or when the work period of injection molding exceeds a set period of time. For example, cleaning is performed approximately once a day. This cleaning causes inconveniences such as an increased burden of the worker and a decrease in the productivity of molded articles.

[0123] Therefore, the injection molding machine **10** according to the embodiment is structured to deliberately mold an article with burrs leaked between the fixed mold **810** and the movable mold **820**, and the cleaning process is performed to suck the debris at the boundaries between the molds (the fixed mold **810** and the movable mold **820**). The injection molding machine **10**, for example, automatically performs the cleaning process after performing injection molding a set number of

times, and removes the debris at the boundaries between the molds. By this means, the injection molding machine **10** reduces the number of times the worker performs cleaning, thus making it possible to reduce the burden of the worker, and improve the productivity of molded articles. [0124] In addition, the injection molding machine **10** may be structured to perform the cleaning process only once between one injection molding and another injection molding, or may be structured to perform the cleaning process multiple times between one injection molding and another injection molding. For example, the injection molding machine **10** may be structured to change the number of times to perform the cleaning process in conjunction with, for example, the number of times normal injection molding is performed. Also, for example, the injection molding machine **10** may be structured to perform the cleaning process at the end of an operation in which injection molding is repeated continuously.

[0125] As shown in FIG. 5A to FIG. 5C, during the cleaning process, the injection molding machine **10** controls the fixed mold **810** and the movable mold **820** to separate from each other and form a gap therebetween such that burrs are formed at the boundaries between the molds. For example, the control device **700** operates the mold thickness adjustment mechanism **180** to adjust (make longer) the interval L between the fixed platen **110** and the toggle support **130**, thereby setting the interval between the fixed platen **110** and the movable platen **120** in injection molding to the interval for the cleaning process. As a consequence of this mold thickness adjustment, when the injection molding machine **10** performs the mold closing step, it becomes possible to dispose the movable mold **820** at a burr-forming position, where the movable mold **820** is separated from the fixed mold **810** further than at the mold clamping position in normal injection molding.

[0126] The upper diagram of FIG. 5C shows the burr-forming position when the molds are closed during the cleaning process, and the lower diagram in FIG. 5C shows, for comparison, the mold clamping position when the molds are clamped in normal injection molding. The burr-forming position is located on the negative X-axis direction side with respect to the mold clamping position. For example, depending on the molding material's fluidity or the like, the burr-forming position may be set at a gap that is formed between the boundaries of the molds and is approximately 0.5 mm to 2 mm. This burr-forming position may be set in advance by performing experiments or simulations, and stored in the control device **700** with the interval for the cleaning process.

Alternatively, the burr-forming position may be set in any desired way based on operations that the worker performs on the operation device **750**.

[0127] In addition, in the cleaning process, the injection molding machine **10** may execute control such that the mold clamping force is reduced, while maintaining mold touch (contact) is maintained, without completely separating the fixed mold **810** and the movable mold **820**. In other words, the burr-forming position may be set as close as possible to the mold clamping position. However, employing a structure that reduces the mold clamping force may result in a situation in which the force that the toggle mechanism **150** can withstand varies depending on the amount by which the movable mold **820** opens from the fixed mold **810**. Therefore, depending on the pressure from the molding material, the movable platen **120** need not be stopped at the desired position. It is therefore more preferable if the injection molding machine **10** performs the cleaning process by controlling the mold clamping force to zero by setting the burr-forming position such that a gap is formed between the fixed mold **810** and the movable mold **820**.

[0128] If the movable mold **820** is moved to the burr-forming position and then the injection device **300** injects the molding material into the mold device **800**, the molding material leaks from the cavity space **801** into the boundaries between the molds. The leaked molding material sucks debris and turns into burrs when cooled down. Therefore, by deliberately molding an article with burrs, the injection molding machine **10** makes it possible to remove the debris from the boundaries between the fixed mold **810** and the movable mold **820**.

[0129] However, during the cleaning process, the injection molding machine **10** must not move (retract) the movable mold **820** that is disposed in the burr-forming position, in the negative X-axis

direction. Assuming that the toggle mechanism **150** is bent significantly in the burr-forming position, if the molding material is injected in the mold device **800** in this state, it is predictable that the burr-forming position cannot be maintained and a significant amount of burrs will be produced. In this case, the injection molding machine **10** will be unable to remove the molded article with burrs from the movable mold **820**. Consequently, by adjusting the thickness of the molds using the mold thickness adjustment mechanism **180**, the injection molding machine **10** according to embodiment provides an area in which the thrust of the crosshead **151** (which is, for example, a force by ball screws and a motor, a force by a linear actuator such as a hydraulic cylinder, etc.) is amplified by the toggle mechanism **150**.

[0130] The area in which the thrust of the crosshead **151** is amplified by the toggle mechanism **150** is where the toggle factor is 1/1 or greater. In other words, in the injection molding machine **10**, the mold thickness adjustment mechanism **180** adjusts the molds' thickness such that the toggle factor of the toggle mechanism **150** is 1/1 or greater at the burr-forming position. The toggle mechanism **150** keeps the toggle factor less than 1/1 in the majority of the molds' opening/closing stroke. Depending on the design of the toggle mechanism **150**, the area in which the amplification by the toggle mechanism **150** applies (that is, the range in which the toggle factor is 1/1 or greater) is approximately 10% of the mold opening/closing stroke.

[0131] However, it is more preferable if the toggle mechanism **150** makes the toggle factor 5 times to 10 times larger. By making the toggle factor 5 times to 10 times larger or more, it then becomes possible to more reliably prevent misplacement of the movable platen **120** and the movable mold **820** during the cleaning process. Next, the toggle factor of the toggle mechanism **150** will be explained with reference to FIG. 6.

[0132] As shown in FIG. 6, the toggle mechanism **150** has a crosshead **151** that moves in mold opening/closing directions, and a pair of link groups that bend and expand following the movement of the crosshead **151**. Each link group includes a first link **152**, a second link **153**, and a third link **154**. One end of the first link **152** is swingably attached to the movable platen **120** by a first pin **155**. One end of the second link **153** is swingably attached to the other end of the first link **152** by a second pin **156**. Also, the other end of the second link **153** is swingably attached to the toggle support **130** by a third pin **157**. One end of the third link **154** is swingably attached to the inside of the second link **153**, at an intermediate position, by a fourth pin **158**. The other end of the third link **154** is swingably attached to a protruding part formed on an outer surface of the crosshead **151** by a fifth pin **159**.

[0133] Given the link groups structured in this manner, by making the crosshead **151** advance in the positive X-axis direction, the third link **154** moves the intermediate position of the second link **153** in the opening direction. One end of the second link **153** moves in the positive X-axis direction, so that the connecting point with the first link **152** (second pin **156**) extends, pushing the first link **152** in the positive X-axis direction. Consequently, following this advancement of the first link **152**, the movable platen **120** advances in the positive X-axis direction. On the other hand, when the crosshead **151** is retracted in the negative X-axis direction, the link group moves the movable platen **120** in the negative X-axis direction, resulting in the reverse of the above operation.

[0134] The toggle mechanism **150** amplifies the driving force of the mold clamping motor **160** and transmits it to the movable platen **120**. The amplification factor through the toggle mechanism **150** is also referred to as "toggle factor." The toggle factor α can be calculated by the following mathematical expression 1:

$$[00001] \quad M = \text{.Math.FTogglefactor} = M / F \quad (\text{MathematicalExpression1})$$

[0135] In the mathematical expression 1, the toggle factor α corresponds to the coefficient part of F, which is the force from the crosshead **151**. M is the force obtained by multiplying the force F of the crosshead **151** by the toggle factor α , and can be applied to the movable platen **120**. In other words, if the toggle factor α is 1/1 or greater, a force M that is greater than or equal to the force F of the

crosshead **151** is applied to the movable platen **120**. Here, the force M , obtained by multiplying the force F from the crosshead **151** by the toggle factor α , can be measured using a detector such as a strain sensor. Also, the toggle factor α can be changed by changing the arrangement of the pins **156** to **159** and changing the intersecting angle of the line segments connecting the pins.

[0136] As described above, the toggle mechanism **150** can amplify the driving force of the mold clamping motor **160** according to the toggle factor α , and transmit it from the crosshead **151** to the movable platen **120**. If the toggle factor α is 1/1 or greater, the toggle mechanism **150** can obtain a fixing force that prevents the movable platen **120** and the fixed mold from retracting, even when the injection device **300** injects the molding material into the mold device **800**. That is, the mold clamping device **100** using the toggle mechanism **150** can easily eliminate the inconvenience of the movable platen **120** opening due to its inability to withstand the pressure when the mold device **10** is filled with the molding material. In particular, the mold thickness adjustment by the mold thickness adjustment mechanism **180** provides an advantage of allowing the injection molding machine **10** to be used at positions where the toggle factor α is high. Although the toggle factor α is increased at burr-forming positions, increasing the toggle factor α entails a disadvantage of increasing the load on each component.

[0137] Also, when the movable mold **820** is moved away from the fixed mold **810**, usually, the toggle factor α decreases, whereas, when the movable mold **820** is brought closer to the fixed mold **810**, the toggle factor α increases. If the toggle factor α is set high, for example, at a position where the movable mold **820** is away from the fixed mold **810** and where the toggle factor α is not needed, the toggle factor α may become too high at the mold clamping position, and the mold clamping force may become too high. Consequently, the toggle factor α is set to less than 1 at positions where the movable mold **820** is away from the fixed mold **810**.

[0138] However, according to this embodiment, the arrangement of the pins **156** to **159** is designed such that the toggle factor α is 1 or greater in burr-forming positions even while the movable mold **820** is away from the fixed mold **810**.

[0139] The structure of the toggle mechanism **150** is by no means limited to the one shown in FIG. 6. For example, while each link group has five nodes in FIG. 6, each link group may have four nodes instead, or one end of the third link **154** may be connected to the node between the first link **152** and the second link **153**.

[0140] Referring back to FIG. 3, the control device **700** has a main control part **711a**, a cleaning control part **711b**, and a switch determining part **711c**, inside the mold clamping control part **711**, for performing normal injection molding and the cleaning process. The injection molding machine **10** can control the operation of the mold clamping device **100** in normal injection molding and the cleaning process, based on the process that each functional part of the mold clamping control part **711** performs.

[0141] To be more specific, the main control part **711a** is a functional part for controlling normal injection molding. The main control part **711a** controls the operation (the mold closing step, the pressure increasing step, the mold clamping step, the pressure releasing step, the mold opening step, etc.) of the mold clamping device **100** in injection molding (see also FIG. 4A). For example, the main control part **711a** controls the mold clamping motor **160** to move the movable platen **120** and the movable mold **820** to the positions set in respective processes (the mold opening/closing start positions, moving speed switching positions, mold closing completion position, mold clamping position, etc.). In addition, the main control part **711a** controls the mold clamping motor **160** by using information detected by the mold clamping motor encoder **161** to adjust the position of the movable mold **820**.

[0142] The cleaning control part **711b** is a functional part for controlling the cleaning process. When switching injection molding to the cleaning process, the cleaning control part **711b** controls the operation of the mold thickness adjustment mechanism **180** of the mold clamping device **100** to switch the positions of the movable platen **120** and the movable mold **820** to the positions for the

cleaning process. The positions for the cleaning process are positions where, during the mold closing step, the movable platen **120** and the movable mold **820** are placed in burr-forming positions, as described earlier.

[0143] After this mold thickness adjustment is performed, the cleaning control part **711b** controls the operation (the mold closing step, the position maintaining step, the mold closing step, etc.) of the mold clamping device **100** in the cleaning process (see also FIG. **4B**). The mold closing step in the cleaning process is a step of moving the movable platen **120** and the movable mold **820** up to burr-forming positions. The position maintaining step in the cleaning process is a step of maintaining the movable mold **820** at the burr-forming position and controlling the injection device **300** to perform the injection step (the filling step and the maintenance step) and the cooling step during this period. The mold opening step in the cleaning process is a step of retracting the movable platen **120** and the movable mold **820** from the burr-forming positions. Note that, to reduce the mold clamping force between the fixed mold **810** and the movable mold **820** while holding these molds in contact with each other during the cleaning process, the cleaning control part **711b** may perform the pressure increasing step between the mold closing step and the position maintaining step, and perform the pressure releasing step between the position maintaining step and the mold opening step.

[0144] The switch determining part **711c** determines whether to switch from normal injection molding to the cleaning process and whether to switch from the cleaning process to normal injection molding. For example, the switch determining part **711c** has a threshold for determining whether or not to start the cleaning process, counts the actual number of times injection molding is performed, and determines to start the cleaning process when this actual number reaches or exceeds the threshold. The same applies when returning from the cleaning process to injection molding. Alternatively, the switch determining part **711c** may report whether the cleaning process can be performed through the display device **760** or the like, and determine to start the cleaning process following the worker's control.

[0145] The injection molding machine **10** according to the embodiment is basically structured as described above. Now, an example of the cleaning process will be described below with reference to FIG. **7** and FIG. **8**.

[0146] The control device **700** of the injection molding machine **10** performs steps **S101** to **S114** in FIG. **7** and FIG. **8**, in order, as an operation method of the cleaning process.

[0147] To be more specific, the switch determining part **711c** of the control device **700** determines whether or not to perform the cleaning process while the main control part **711a** is performing normal injection molding (step **S101**). For example, the switch determining part **711c** increments the actual number of times injection molding is performed at the start of the mold closing step by the mold clamping device **100**, compares this value against the starting threshold, and determines whether or not to switch to the cleaning process after injection molding is performed. When the control device **700** determines to perform the cleaning process (step **S101**: YES), the flow proceeds to step **S102**, and, when the control device **700** determines not to perform the cleaning process (step **S101**: NO), step **S101** is repeated.

[0148] Then, while the control device **700** is performing normal injection molding, the control device **700** prepares for the next cleaning process. To be more specific, in normal injection molding, the control device **700** performs the injection step (the filling step and the maintenance step) by the injection device **300** while performing the pressure increasing step and the mold clamping step by the mold clamping device **100** (step **S102**).

[0149] Then, while the cooling step is being performed after the injection step, the measurement control part **714** of the control device **700** performs a measurement step that suits the following cleaning process, and measures the amount of molding material in this measurement step, envisaging that burrs will be formed (step **S103**). The amount of molding material taking into account the burrs that will be formed depends, for example, on the shape of the molded article and

the molding conditions. For example, the amount of molding material may be adjusted to increase by approximately 1% to 10% compared to the amount of molding material in normal injection molding. Alternatively, the amount of molding material need not be different between normal injection molding and the cleaning process.

[0150] After the cooling step is performed, the control device **700** controls the mold clamping device **100** to perform the pressure releasing step and the mold opening step for normal injection molding, and also controls the ejector device **200** to perform the ejection step of ejecting the molded article (step **S104**). However, after the molded article is ejected, the control device **700** may stop the ejector rods **210** at a midway point, rather than retracting the ejector rods **210** to their return limit. By stopping the ejector rods **210** at a midway point, a gap can be formed in parts that the ejector rods **210** push when the cleaning process is performed, so that the release of the molded article having burrs can be promoted. The parts that the ejector rods **210** may preferably be parts of burrs, for example.

[0151] When the ejector device **200** is retracted, the normal injection molding cycle ends. Then, the control device **700** transitions to a molding cycle in which the cleaning process is performed. In this case, the cleaning control part **711b** of the control device **700** performs mold thickness adjustment by operating the mold thickness adjustment mechanism **180** such that the movable platen **120** and the movable mold **820** are retracted back to the positions for the cleaning process, as a preparation for the cleaning process (step **S105**). For example, the cleaning control part **711b** operates the mold thickness adjustment mechanism **180** such that the movable mold **820** is spaced 1.5 mm apart from the fixed mold **810** in the mold closing step of the cleaning process.

[0152] When the mold thickness adjustment by the mold thickness adjustment mechanism **180** is completed, the cleaning control part **711b** executes a molding cycle in which the cleaning process is performed. As shown in FIG. 8, in the cleaning process, the cleaning control part **711b** first controls the mold clamping device **100** to move the movable platen **120** and the movable mold **820** in the positive X-axis direction to perform the mold closing step (step **S106**). By this means, the movable platen **120** and the movable mold **820** are brought close to the fixed mold **810** and disposed in burr-forming positions (see also FIG. 5C). These burr-forming positions are, for example, positions where the boundaries of the fixed mold **810** and the movable mold **820** are spaced apart by a gap of 1.5 mm.

[0153] The cleaning control part **711b** then controls the mold clamping device **100** to perform the position maintaining step, while the injection control part **713** controls the injection device **300** to perform the injection step (the filling step and the maintenance step) (step **S107**). In this position maintaining step, the cleaning control part **711b** drives the mold clamping motor **160** to maintain the toggle factor α of the toggle mechanism **150** at 1/1 or greater without changing the burr-forming positions. By this means, even if the injection device **300** fills the mold device **800** with the molding material in the injection step, the movable platen **120** and the movable mold **820** can be prevented from being retracted (from being misplaced).

[0154] After the injection step is performed, the control device **700** moves on to the cooling step and finishes the molding the article with burrs (step **S108**). The burrs have debris attached thereto, from the boundaries between the molds.

[0155] Furthermore, the control device **700** performs the mold opening step, retracts the movable mold **820** from the fixed mold **810**, and controls the ejector device **200** to perform the step of ejecting the molded article (step **S109**). By this means, the molded article is removed from the movable mold **820**, and the debris at the boundaries between the molds are removed. The molded article with burrs is removed and discarded, for example, by an automatic sorting machine or the worker. Note that the worker may perform the task of removing the burrs from the molded articles with burrs.

[0156] Also, the switch determining part **711c** determines whether or not to resume normal injection molding during the current cleaning process (step **S110**). For example, the switch

determining part **711c** monitors whether or not the cleaning process has been performed a set number of times. If the cleaning process has been performed the set number of times, the switch determining part **711c** determines to resume injection molding. On the other hand, if the cleaning process has not been performed the set number of times, the switch determining part **711c** determines to repeat the cleaning process. Note that in FIG. 8, the decision to resume injection molding is described as being made after the ejection step is performed, but the decision to resume injection molding may be made in parallel with the injection step of the cleaning process, and is preferably made at least before the cooling retraction in the cleaning process. By this means, when the measurement step is performed during the cooling step, the amount of molding material can be measured appropriately in accordance with the next injection molding.

[0157] If normal injection molding is to be resumed (step **S110**: YES), the flow proceeds to step **S111**, and the molding material for injection molding is measured. Then, after the ejection step is performed, the control device **700** switches to the molding cycle in which the next normal injection molding will be performed. In this case, the main control part **711a** of the control device **700** operates the mold thickness adjustment mechanism **180** to advance the movable platen **120** and the movable mold **820** to the positions for injection molding in preparation for normal injection molding (step **S112**).

[0158] After step **S112**, the injection molding machine **10** resumes normal injection molding (step **S113**). As a consequence of the preparatory mold thickness adjustment, the movable platen **120** and the movable mold **820** resume their original positions, so that the main control part **711a** can operate the mold clamping device **100** can stably perform the mold closing step, pressure increasing step, mold clamping step, pressure releasing step, mold opening step, and so on.

[0159] On the other hand, if the cleaning process is to be repeated (step **S110**: NO), the flow proceeds to step **S114** and the molding material for the cleaning process is measured. After the ejection step is performed, the control device **700** returns to step **S106** and performs the cleaning process again.

[0160] As described earlier, the injection molding machine **10** according to the embodiment employs a simple structure in which the position of the movable mold **820** is adjusted under the control of the control device **700**, thereby enabling an article to be molded with burrs and the boundaries between the molds to be cleaned up in an efficient manner. Moreover, the movable platen **120** and the movable mold **820** disposed in burr-forming positions can stably maintain the burr-forming positions because the toggle factor α of the toggle mechanism **150** is $1/1$ or greater. This allows the shape (size, etc.) of the burrs to be adjusted appropriately to remove dust. As a result of this, the injection molding machine **10** can reduce the number of times the worker has to clean up the mold device **800**, thereby reducing the burden of the worker and improving the overall productivity of molded articles. For example, the injection molding machine **10** can change the number of times the worker does the cleaning of the mold device **800** from once a day to once every several days (for example, once every seven days).

[0161] In particular, during the cleaning process, the injection molding machine **10** can stably mold burrs of a set size by forming a gap between the movable mold **820** and the fixed mold **810** that have been moved to burr-forming positions.

[0162] Also, before performing the cleaning process, the injection molding machine **10** operates the mold thickness adjustment mechanism **180** to adjust the interval between the fixed platen **110** and the movable platen **120** to an interval for the cleaning process. By this means, the injection molding machine **10** can stably maintain the toggle factor α at a high magnification, even when the movable platen **120** and the movable mold **820** are placed in burr-forming positions during the cleaning process.

[0163] Furthermore, after the cleaning process is completed, the injection molding machine **10** can adjust the interval between the fixed platen **110** and the movable platen **120** to an interval for injection molding and resume injection molding, thereby allowing injection molding to be resumed

smoothly after the cleaning process is performed.

[0164] Also, by repeating the cleaning process multiple times, the injection molding machine **10** can sufficiently remove the debris produced at the boundaries between the fixed mold **810** and the movable mold **820**.

[0165] Also, the injection molding machine **10** determines the start and/or end of the cleaning process prior to the measurement step of the injection device **300**, allowing it to measure an appropriate amount of molding material in the measurement step. By this means, the control device **700** can more smoothly switch between normal injection molding and the cleaning process.

[0166] Also, by setting the toggle factor α of the toggle mechanism **150** to 5 times larger or more at burr-forming positions, it is possible to more reliably prevent the movable platen **120** that has been moved to the burr-forming position from being misplaced.

[0167] Note that the injection molding machine **10** according to embodiment is by no means limited to the above embodiment and may be provided in a variety of altered ways. For example, the injection molding machine **10** has only to be disposed at a burr-forming position where the toggle factor α is 1/1 or greater, and the method of adjusting the positions of the movable platen **120** and the movable mold **820** is by no means limited to the mold thickness adjustment by the mold thickness adjustment mechanism **180**. For example, the injection molding machine **10** may position the movable platen **120** and the movable mold **820** at burr-forming positions by controlling the mold clamping motor **160** without performing mold thickness adjustment. Even in this case, if the toggle factor α is 1/1 or greater, it is still possible to prevent or substantially prevent the movable platen **120** and the movable mold **820** from being retracted.

[0168] Also, for example, the injection molding machine **10** may employ a method in which filling of the molding material is started with a wide gap as in compression molding, and, after the filling is done, the movable platen **120** and the movable mold **820** are moved to crush the molded article and form burrs. In this case, even if the movable platen **120** and the movable mold **820** are placed in burr-forming positions, the pressure of the molding material is low, and burrs can be formed while increasing the toggle factor α by moving the movable platen **120**.

[0169] The injection molding machine **10** according to the embodiment disclosed herein is illustrative in all respects and is not restrictive. The embodiment can be modified and improved in various forms without departing from the scope and spirit of the appended claims. The matters described in the above embodiment can have other structures to the extent that they are not inconsistent, and can also be combined to the extent that they are not inconsistent.

Claims

1. An injection molding machine comprising: a mold clamping device including: a movable platen having a movable mold; a fixed platen having a fixed mold that faces the movable mold; and a toggle mechanism configured to move the movable platen in directions in which a mold device opens and closes, relative to the fixed platen, and dispose the movable platen in a mold clamping position; an injection device configured to inject a molding material into a cavity space formed between the movable mold disposed in the mold clamping position and the fixed mold; and a control device configured to control operation of the injection device and the mold clamping device, wherein the control device is configured to perform a cleaning process, in which, while the movable platen is at a burr-forming position where the movable mold is farther away from the fixed mold than at the mold clamping position, the injection device injects the molding material into the mold device to mold an article with burrs, and wherein the toggle mechanism is configured such that a toggle factor at the burr-forming position is a magnification factor of 1 or greater.
2. The injection molding machine according to claim 1, wherein the control device is configured to form, during the cleaning process, a gap between the movable mold having moved to the burr-forming position, and the fixed mold.

3. The injection molding machine according to claim 2, wherein the mold clamping device has a mold thickness adjustment mechanism configured to adjust an interval between the fixed platen and the movable platen, and wherein the control device is configured to operate, before the cleaning process is performed, the mold thickness adjustment mechanism to adjust the interval between the fixed platen and the movable platen to an interval for the cleaning process.
 4. The injection molding machine according to claim 3, wherein the control device is configured to adjust the interval between the fixed platen and the movable platen to an interval for injection molding and resumes injection molding after the cleaning process is performed.
 5. The injection molding machine according to claim 1, wherein the control device is configured to repeat the cleaning process a plurality of times.
 6. The injection molding machine according to claim 5, wherein the control device is configured to perform a detection of a start of the cleaning process or an end of the cleaning process, or both, before the injection device performs a measurement step, and wherein the control device is configured to perform the measurement step, the measurement step including measuring an amount of the molding material according to a result of the detection.
 7. The injection molding machine according to claim 1, wherein the toggle mechanism is configured such that the toggle factor at the burr-forming position is a magnification factor of 5 or greater.
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